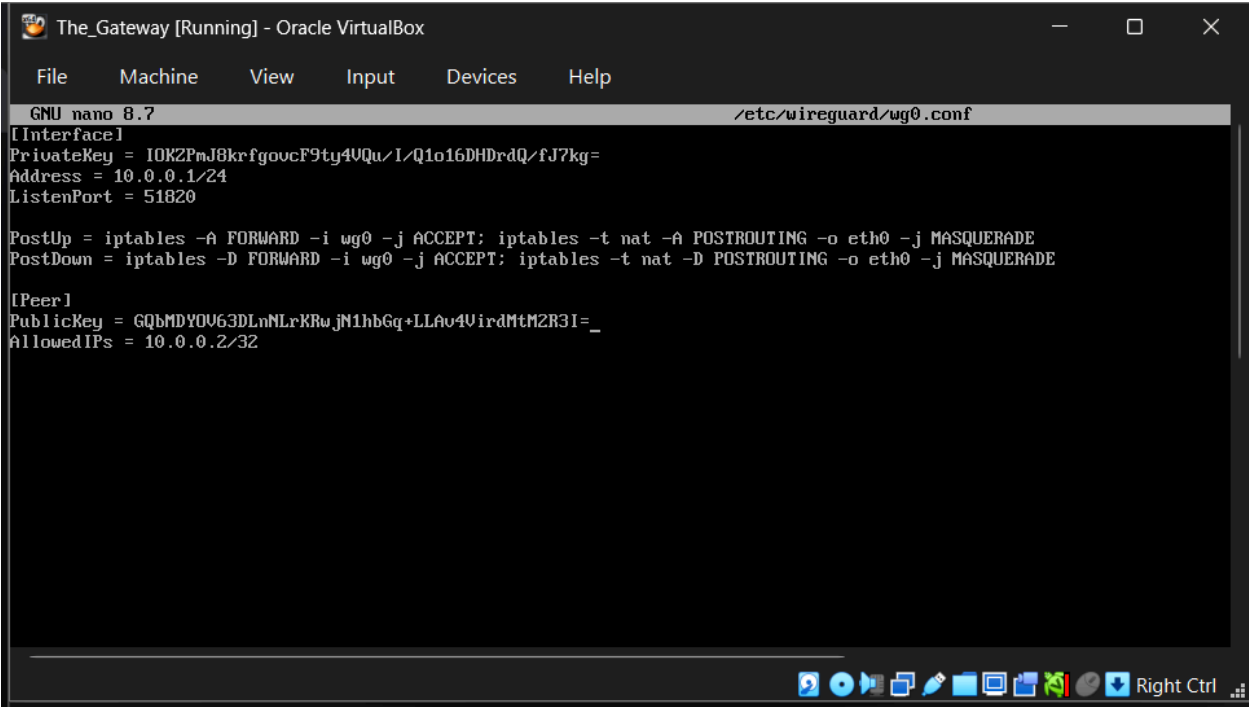
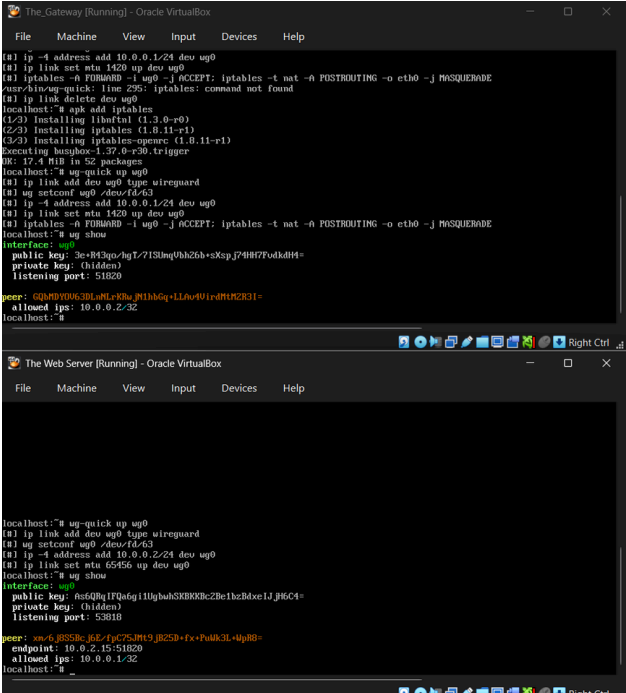


Deploying Modern VPNs with WireGuard

Latency connection	WireGuard (Internal Lab)	OpenVPN (Internal Lab)	Why the Difference?
Average Latency	~0.25 ms to 0.5 ms	~1.2 ms to 2.5 ms	WireGuard's code is extremely lean and processed directly by the CPU at a higher priority.
CPU Overhead	Very Low	Moderate to High	OpenVPN uses heavier encryption libraries (OpenSSL) that require more processing power per packet.
Initial Connection	Instant (< 100ms)	2–5 seconds	WireGuard has no complex "handshake negotiation" phase; it just starts sending data.

Cryptokey Routing replaces a Certificate Authority (CA) by mapping unique Public Keys directly to specific internal IP addresses within the configuration file. In traditional VPNs, a CA must sign digital certificates to "vouch" for a user's identity. In WireGuard, the Public Key itself is the identity. By hard-coding which key is allowed to use which IP, the system ensures that only the holder of the matching Private Key can communicate.

This method eliminates the need for a complex "trust hierarchy" and the management of .crt or .pem files. Because the Gateway simply drops any packet that doesn't come from a recognized key-IP pair, the possession of the key serves as both the credential and the routing instruction. This makes the network significantly leaner and more secure than managing a traditional certificate infrastructure.



The_Gateway [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
Executing bash-5.3.3-r1.post-install
( 8/21) Installing openresolv (3.17.4-r0)
( 9/21) Installing zstd-libs (1.5.7-r2)
(10/21) Installing libelf (0.194-r0)
(11/21) Installing libmnl (1.0.5-r2)
(12/21) Installing iproute2-minimal (6.17.0-r0)
(13/21) Installing ifupdown-ng-iproute2 (0.12.1-r7)
(14/21) Installing libxtables (1.8.11-r1)
(15/21) Installing iproute2-tc (6.17.0-r0)
(16/21) Installing iproute2-ss (6.17.0-r0)
(17/21) Installing iproute2 (6.17.0-r0)
Executing iproute2-6.17.0-r0.post-install
(18/21) Installing wireguard-tools-wg-quick (1.0.20250521-r1)
(19/21) Installing ifupdown-ng-wireguard-quick (0.12.1-r7)
(20/21) Installing wireguard-tools (1.0.20250521-r1)
(21/21) Installing wireguard-tools-openrc (1.0.20250521-r1)
Executing wireguard-tools-openrc-1.0.20250521-r1.post-install
*
* To use the WireGuard OpenRC script, you need to create a symbolic link to it with the configuration name:
*   ln -s /etc/init.d/wg-quick /etc/init.d/wg-quick.wg0
* And then call it instead:
*   rc-service wg-quick.wg0 start
*
Executing busybox-1.37.0-r30.trigger
OK: 15.4 MiB in 49 packages
localhost:~# _
```

Right Ctrl